107.330 Statistical Simulation and Computerintensive Methods - Assignment 1

Exercises 1

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Instructions

For each exercise:

- reason about the numerical behaviour before running the code;
- compare alternative implementations where appropriate;
- explain *why* the results differ when they do;
- distinguish statistical issues from numerical issues;
- clearly document your reasoning and R code.

You may use base R and the functions introduced in the lecture.

Submission

Solutions must be submitted in **TUWEL no later than 18.11.2026 at 23:59**.

Exercise points

Exercise points are awarded only in connection with the corresponding exercise session.

To receive the point for an exercise, you must submit your solution by the deadline.

Use of generative AI

The use of **generative AI tools to solve the exercise tasks is not permitted**.

The purpose of the exercises is to develop your own ability to translate statistical ideas into algorithms, implement them in R, investigate numerical behaviour, and explain your reasoning.

The submitted solutions, including the reasoning and R code, must therefore reflect **your own work**.

Exercise 1 — Variance after a change of origin

Suppose a data set is given by

$$x_i = a + z_i,$$

where a is a constant and z_i is a sample with moderate variability.

Mathematically,

$$\mathrm{Var}(X + a) = \mathrm{Var}(X).$$

Consider the following experiment:

```
set.seed(1)

z <- rnorm(100000, mean = 0, sd = 3)

offsets <- c(0, 1e4, 1e8, 1e12, 1e14)
```

Tasks

1. For each value of `a` in `offsets`, form `x <- a + z`.
2. Compute the sample variance using:
   - `var(x)`;
   - the two-pass algorithm;
   - the one-pass computational formula

$$s^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}.$$

3. Put the results in a data frame with one row per offset.
4. Compare each result with `var(z)`. At approximately what size of a does the one-pass formula begin to lose useful accuracy?
5. Explain why adding a constant leaves the variance unchanged mathematically but can still affect a particular numerical algorithm.
6. Suppose you receive measurements in nanometres with values around 10^{12}, but the scientifically relevant variation is only a few units. Would you preprocess the data before computing descriptive statistics? If yes, what transformation would you use and why?

Exercise 2 — A probability that disappears

Let $Z \sim N(0, 1)$. For large x,

$$P(Z > x)$$

is very small.

Consider the following values:

```r
x <- c(5, 8, 10, 20, 30, 40)
```

Tasks

1. Compute the upper-tail probability in each of the following ways:

```r
1 - pnorm(x)
pnorm(x, lower.tail = FALSE)
pnorm(x, lower.tail = FALSE, log.p = TRUE)
```

2. Identify where the first expression begins to give a misleading answer.
3. Explain why `1 - pnorm(x)` can fail even when `pnorm(x)` itself is accurately computed.
4. For each x, recover a probability from the log-probability using `exp()`. At what point does even that conversion become unhelpful?
5. Imagine a multiple-testing problem with a test statistic corresponding to $Z = 9$. You need the p-value to compare it with a very stringent threshold. Which representation of the probability would you carry through the computation, and why?

Exercise 3 — Comparing two models without computing their likelihoods

Suppose we observe independent Bernoulli data

```r
set.seed(2)
y <- rbinom(5000, size = 1, prob = 0.03)

p1 <- 0.03
p2 <- 0.035
```

For a Bernoulli model with success probability p,

$$L(p) = \prod_{i=1}^{n} p^{y_i}(1-p)^{1-y_i}.$$

Tasks

1. Compute `L(p1)` and `L(p2)` directly in R by multiplying the individual probabilities.
2. Compute the corresponding log-likelihoods

$$\ell(p) = \sum_{i=1}^{n} \left[y_i \log p + (1-y_i) \log(1-p) \right].$$

3. Which model is preferred according to the direct likelihoods? Which is preferred according to the log-likelihoods?

4. Compute the log-likelihood difference

$$\ell(p_1) - \ell(p_2).$$

Explain why this quantity can be more useful than either likelihood itself.

5. Compute the likelihood ratio

$$\frac{L(p_1)}{L(p_2)}$$

in a numerically stable way using the log-likelihoods.

6. Why is this issue especially important in Bayesian computation, maximum likelihood estimation, and sequential analysis?

Exercise 4 — Diagnosing a numerically suspicious covariance matrix

Consider two variables that are almost linear copies of each other:

```r
set.seed(3)

n <- 5000
u <- rnorm(n)

x1 <- 1e8 + u
x2 <- 1e8 + u + rnorm(n, sd = 1e-4)

X <- cbind(x1, x2)
```

Tasks

1. Compute the covariance matrix using `cov(X)`.
2. Compute the correlation between `x1` and `x2`.
3. Compute the eigenvalues of the covariance matrix.
4. Now implement covariance using the algebraic one-pass identity

$$\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y),$$

with sample corrections as needed.

5. Compare the resulting covariance matrix with `cov(X)`.
6. Repeat the calculation after centring both variables first:

```r
Xc <- scale(X, center = TRUE, scale = FALSE)
```

7. A statistical procedure requires inversion of the covariance matrix. What warning signs in this example would make you cautious about doing that directly? Suggest at least two safer approaches.

Exercise 5 — Whitening is not unique

Let $X \in \mathbb{R}^{n \times p}$ be a data matrix whose rows are observations. After centring the data, let

$$S = \mathrm{Cov}(X)$$

denote the sample covariance matrix.

A matrix W is a **whitening matrix** if the transformed observations

$$Z = XW^\top$$

have identity covariance:

$$\mathrm{Cov}(Z) = WSW^\top = I_p.$$

The goal of this exercise is to investigate why whitening is **not unique** and how different matrix decompositions lead to different, but valid, whitening transformations.

Tasks

1. Simulate $n = 1000$ observations from a 5-dimensional multivariate normal distribution with a non-diagonal covariance matrix. You may construct such data without an additional package by starting with independent standard normal variables and multiplying them by a suitable matrix.

   Centre the resulting data and inspect its sample covariance matrix S.

2. Compute the eigendecomposition

   $$S = Q\Lambda Q^\top.$$

   Use it to construct the **PCA whitening matrix**

   $$W_{\mathrm{PCA}} = \Lambda^{-1/2} Q^\top.$$

   Apply the transformation and verify numerically that

   $$W_{\mathrm{PCA}} S W_{\mathrm{PCA}}^\top \approx I.$$

3. Construct the **ZCA whitening matrix**

   $$W_{\mathrm{ZCA}} = Q\Lambda^{-1/2} Q^\top.$$

   Again verify that the transformed data have approximately identity covariance.

4. Compute a Cholesky factorization

   $$S = LL^\top.$$

   Construct a whitening transformation based on L^{-1}, and verify that it also produces identity covariance.

   Is the Cholesky factor L a matrix square root of S in the sense that

   $$L^2 = S?$$

   Explain carefully.

5. Compare the three whitened data sets.

   - Are the transformed observations identical?
   - Do they have the same covariance matrix?
   - Which aspects of the data have changed differently?

6. Show mathematically that if W is a whitening matrix and R is any orthogonal matrix, then

   $$RW$$

   is also a whitening matrix.

   Use this result to explain why whitening is not unique.

7. Now create a more difficult data set by making the fifth variable almost a linear combination of another variable. For example,

```
X[, 5] <- X[, 1] + 1e-7 * rnorm(n)
```

Re-centre the data and investigate:

- the eigenvalues of S;
- `kappa(S)`;
- the three whitening transformations;
- the covariance matrices after whitening.

What numerical difficulties do you observe?

8. Suppose one eigenvalue of S is numerically zero. Suggest at least two reasonable ways to proceed if whitening is still required. Discuss both the numerical and statistical consequences.